Lecture 13

18 February 2026 20:31

Recap

Unconstrained optimization: $f: \mathbb{R}^n \to \mathbb{R}$

$$\min_{x \in \mathbb{R}^n} f(x)$$

Goal: Obtain a x^* such that $\cancel{\nabla f(x^*)} = 0$

Initialize: $x_0 \in \mathbb{R}^n$

Iterate: for $k \geq 0$, $x_{t+1} = x_t + \alpha_t d_t$ where $\alpha_t > 0$ and $\|d_t\| = 1$

- α_t is called a step-size
- d_t is called a direction
- The algorithm is called a descent algorithm if $f(x_{t+1}) \leq f(x_t)$ for all $t \geq 0$

Direction selection:

- If d_t is chosen such that [Equation] for all [Equation] then the algorithm is called a gradient descent algorithm
- If [Equation] for all [Equation] then the algorithm is called a steepest descent algorithm

Steepest descent

$$x_{t+1} = x_t + \alpha_t \underbrace{\left(- \frac{\nabla f(x_t)}{\|\nabla f(x_t)\|} \right)}_{=: d_t \ (\|d_t\| = 1)}$$

Step-size selection:

- Fixed step-size: $\alpha_t = \alpha$

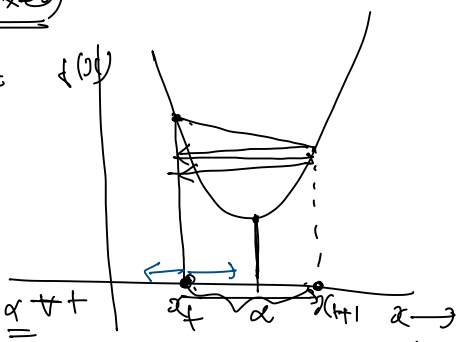
In general $x_{t+1} = x_t + \alpha d_t$ $\quad \|x_{t+1} - x_t\| = \alpha \|d_t\| = \alpha$ (fixed)

  - if [Equation] then, customarily, fixed step-size refers to [Equation] for all [Equation]

$$x_{t+1} = x_t - \alpha \nabla f(x_t)$$ steepest gradient descent with fixed step size.

$$= x_t + \underbrace{\alpha \|\nabla f(x_t)\|}_{\text{step size (not fixed)}} \underbrace{\left(- \frac{\nabla f(x_t)}{\|\nabla f(x_t)\|} \right)}_{\text{direction}}$$

Issue with fixed step size



$x_{t+1} = x_t + \alpha_t d_t$,

$|d_t| = 1$.

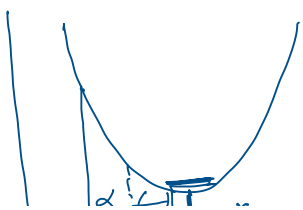
Suppose $\alpha_t = \alpha \ \forall t$

$|x_{t+1} - x_t| = \alpha \underbrace{|d_t|}_{=1} = \alpha$

$d_t \in \{1, -1\}$ have $d_t = 1$

$d_{t+1} = -1$

— we oscillate between x_t & x_{t+1}



use smaller α now

$x_t \quad x_{t+1} \quad x_{t+2} \quad x^*$

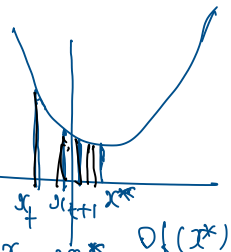
$d_t = 1 \quad d_{t+1} =$

using fixed α leads to oscillations, when we are close to x^*

In case of steepest descent

$$x_{t+1} = x_t - \alpha \nabla f(x_t)$$

$$|x_{t+1} - x_t| = \alpha |\nabla f(x_t)|$$



reduce as $x_t \to x^*$ $\quad \nabla f(x^*) = 0$

- there is reduced chance of fluctuation if α is chosen carefully.

- we could choose very small α
  - it would avoid oscillations until we are very close to x^*
  - but it leads to slow convergence to x^*

Exact line search

$$x_{t+1} = x_t + \alpha_t d_t$$

choose α_t s.t. $f(x_{t+1})$ has the least possible value.

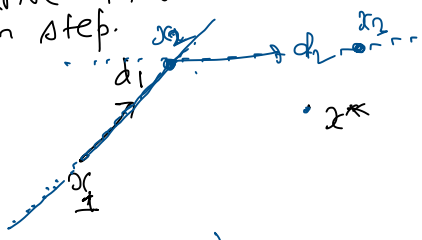
$$\alpha_t = \underset{\alpha}{\operatorname{argmin}} f(x_t + \alpha d_t)$$

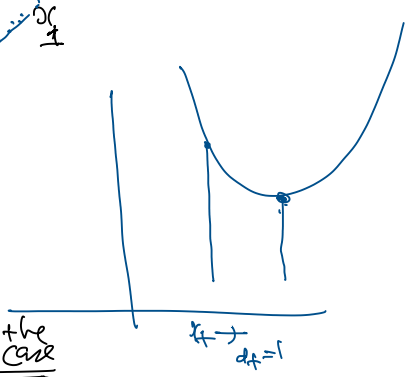
- variable is α

we can set $d_t = -\nabla f(x_t) \ \forall t$

or, any d_t s.t. $d_t^T \nabla f(x_t) < 0$

- we must solve an unconstrained scalar minimization at each step.





f convex

$\nabla f(x) = \alpha (x - x^*)$ need not be the case

- Exact line search: [Equation]
  - Requires solving a "scalar" optimization problem at each step

Armijo's rule

$\alpha_0 = \alpha$

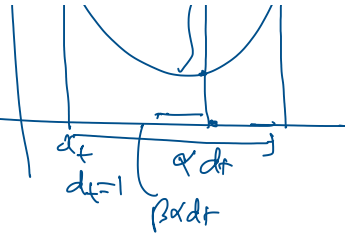
- go to "$x_t + \alpha d_t$"

– is this a "good point"?

if not reduce α to $\beta\alpha$ $\beta \in (0,1)$

is "$x_t + \alpha\beta d_t$" a good point

if not reduce α to $\beta^2\alpha$



if yes $x_{t+1} = x_t + \beta^m \alpha d_t$

"good point"

$$f(x_t + \beta^m \alpha d_t) \le f(x_t) + \sigma (\beta^m \alpha d_t)^T \nabla f(x_t)$$

$\sigma \in (0,1)$

$\alpha \ge 0.5$

First order approx

$$f(x_t + \beta^m \alpha d_t) \approx f(x_t) + \nabla f(x_t)^T (\beta^m \alpha d_t)$$

start with α

if $f(x_t + \alpha d_t) \le f(x_t) + \sigma (\alpha d_t)^T \nabla f(x_t)$

$\Rightarrow x_{t+1} = x_t + \alpha d_t$

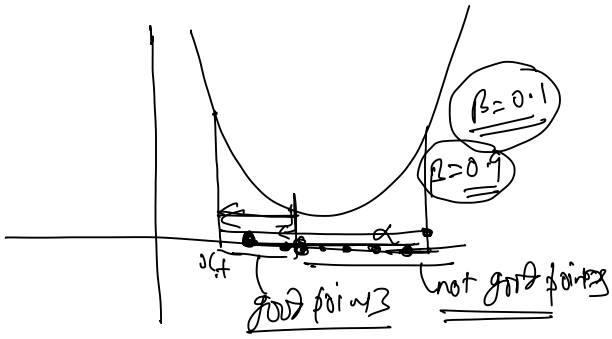
else $f(x_t + \beta\alpha d_t) \le f(x_t) + \sigma (\beta\alpha d_t)^T \nabla f(x_t)$

$\Rightarrow x_{t+1} = x_t + \beta\alpha d_t$

else $f(x_t + \beta^2\alpha d_t) \le f(x_t) + \sigma (\beta^2\alpha d_t)^T \nabla f(x_t)$

$\Rightarrow$

else



- Backtracking line search (Armijo's rule): Fix [Equation] and [Equation] . Given [Equation]

  for $t = 1, 2, 3, \ldots$

  Initialize: [Equation]

  While [Equation]

  [Equation]

  End.

  $x_{t+1} = x_t + \alpha_t d_t$

  end

  $\|x_{t+1} - x_t\| = \alpha_t \|d_t\|$

  - if [Equation] then Armijo's rule converges quicky but the gradient descent algorithm's progress may be slow; if [Equation] is close to [Equation] then Armijo's rule takes several iterations to converge but the overall progress may be substantial

Properties

Fixed step-size rule (with steepest decent direction):
Initialize: [Equation]
Iterate: for [Equation] where [Equation]

For [Equation] ,

[Equation]

Suppose f is Lipschitz continuous with para. B.

$$\|f(x)-f(y)\| \le B\|x-y\|$$

In this case, $\|\nabla f(x)\| \le B \quad \forall x$

$$\Rightarrow \text{R.H.S} \le \frac{\alpha}{2}\sum_{t=0}^{T-1} B^2 + \frac{1}{2\alpha}\|x_0 - x^*\|^2$$

$$= \frac{B^2\alpha T}{2} + \frac{\|x_0 - x^*\|^2}{2\alpha}$$

Let us choose α s.t. RHS is minimized.

$$g(\alpha) = \frac{B^2\alpha T}{2} + \frac{\|x_0 - x^*\|^2}{2\alpha}$$

$$g'(\alpha) = 0 \Rightarrow \frac{B^2 T}{2} - \frac{\|x_0 - x^*\|^2}{2\alpha^2} = 0$$

$$\Rightarrow \alpha = \frac{\|x_0 - x^*\|}{B\sqrt{T}}$$

Substituting in back

$$\sum_{t=0}^{T-1}(\nabla f(x_t))^T(x_t - x^*) \le \frac{B^2\|x_0 - x^*\|^T}{2B\sqrt{T}} + \frac{B\sqrt{T}}{2\|x_0 - x^*\|}\cdot\|x_0 - x^*\|^2$$

$$= \|x_0 - x^*\| B\sqrt{T}$$

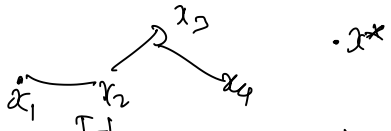
$$\Rightarrow \frac{1}{T}\sum_{t=0}^{T-1}(\nabla f(x_t))^T(x_t - x^*) \le \frac{\|x_0 - x^*\| B}{\sqrt{T}}$$

Suppose f is also convex.

$$f(x^*) \ge f(x_t) + \nabla f(x_t)^T(x^* - x_t)$$

$$\Rightarrow \nabla f(x_t)^T(x_t - x^*) \ge f(x_t) - f(x^*)$$

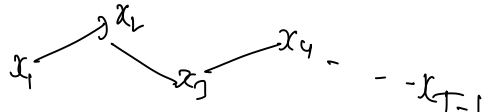
$$\Rightarrow \frac{1}{T}\sum_{t=0}^{T-1}(f(x_t) - f(x^*)) \le \frac{\|x_0 - x^*\| B}{\sqrt{T}}$$



$$\min_{a_t} \le \frac{1}{T}\sum_{t=0}^{T} a_t$$

$$\min_{t=0,\dots,T-1}(f(x_t) - f(x^*)) \le \frac{1}{T}\sum_{t=0}^{T-1}(f(x_t) - f(x^*))$$

$$\Rightarrow \min_{t=0,\dots,T-1}(f(x_t) - f(x^*)) \le \frac{\|x_0 - x^*\| B}{\sqrt{T}}$$



At at each t, store $\underset{0,\dots,t}{\operatorname{argmin}} f(a_t)$

Theorem: (a) If [Equation] is Lipschitz continuous with parameter [Equation] then using [Equation]
[Equation]
(b) If [Equation] is also convex then
[Equation]

* How do we pick $\alpha = \frac{\|x_0 - x^*\|}{B\sqrt{T}}$ when we don't x^*?

we must also know the number of iterations apriori.

Suppose f has Lipschitz CIG gradient with parameter L.

$$\|\nabla f(x) - \nabla f(y)\| \le L\|x-y\| \quad \forall x, y.$$

Descent lemma

$$f(x_{t+1}) \le f(x_t) + \nabla f(x_t)^T (x_{t+1} - x_t) + \frac{L}{2}\|x_{t+1} - x_t\|^2$$

Assume $\alpha = \eta/L$ where $\eta > 0$

$$x_{t+1} = x_t - \left(\frac{\eta}{L}\right)\nabla f(x_t)$$

$$f(x_{t+1}) \le f(x_t) - \frac{\eta}{L}\|\nabla f(x_t)\|^2 + \frac{L}{2}\frac{\eta^2}{L^2}\|\nabla f(x_t)\|^2$$

$$= f(x_t) - \underbrace{\frac{\eta}{L}\left(1 - \frac{\eta}{2}\right)}_{\text{if } \eta < 2}\|\nabla f(x_t)\|^2$$

$$\Rightarrow f(x_t) - f(x_{t+1}) \ge \underbrace{\frac{\eta}{L}\left(1 - \frac{\eta}{2}\right)}_{>0}\|\nabla f(x_t)\|^2$$

sum over $t = 0, \dots, T-1$

$$\Rightarrow \sum_{t=0}^{T-1}(f(x_t) - f(x_{t+1})) \ge \frac{\eta}{L}\left(1 - \frac{\eta}{2}\right)\sum_{t=0}^{T-1}\|\nabla f(x_t)\|^2$$

$$\Rightarrow f(x_0) - f(x_T) \ge \frac{\eta}{L}\left(1 - \frac{\eta}{2}\right)\sum_{t=0}^{T-1}\|\nabla f(x_t)\|^2$$

$\min_x f(x)$

$f(x) \ge f(x^*) \quad \forall x$

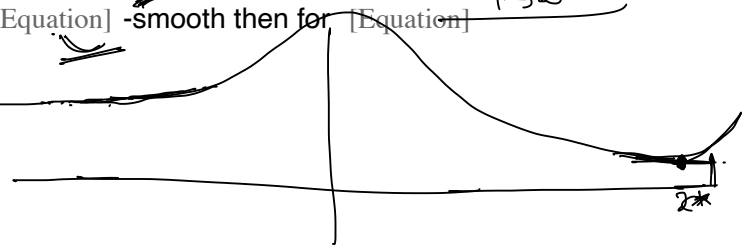
$$\Rightarrow \sum_{t=0}^{T-1}\|\nabla f(x_t)\|^2 \le \frac{f(x_0) - f(x_T)}{\frac{\eta}{L}\left(1 - \frac{\eta}{2}\right)}$$

$$\Rightarrow \lim_{t\to\infty}\|\nabla f(x_t)\| = 0$$

$$\sum_{i=1}^{n}|a_i| \le B \quad \forall n$$

$$\lim_{i\to\infty}|a_i| = 0$$

Theorem: (a) If [Equation] is [Equation] -smooth then for [Equation]
,
[Equation]



(b) If [Equation] is also convex then using [Equation] ,
[Equation]

Theorem: If [Equation] is [Equation] -smooth and strongly convex with parameter [Equation] then using [Equation] ,

1. [Equation] ,
2. [Equation] .

$(m < L)$

$= \Rightarrow \frac{m}{L} < 1$

$1 \Rightarrow 2$

$$f(x_T) \le f(x^*) + \underbrace{\nabla f(x^*)}_{=0}(x_T - x^*) + \frac{L}{2}\|x_T - x^*\|^2$$

$$\le f(x^*) + \frac{L}{2}\left(1 - \frac{m}{L}\right)^T \|x_0 - x^*\|^2$$

Earlier $f(x_T) - f(x^*) = O(1/T)$

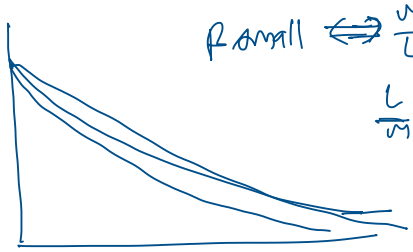
Now $f(x_T) - f(x^*) = O(\beta^T)$

$\beta = \left(1 - \frac{m}{L}\right)$

β small $\Leftrightarrow \frac{m}{L}$ close to 1

$\frac{L}{m}$ condition number

Eventually, much faster convergence



Example $f(x) = \frac{1}{2} x^T Q x + b^T x$

$\nabla f(x) = Qx$

$\nabla^2 f(x) = Q$

f is L-smooth with $L = \lambda_{max}(Q)$

f is strongly convex with $m = \lambda_{min}(Q)$ (if Q is positive definite)

condition number $= \frac{\lambda_{max}(Q)}{\lambda_{min}(Q)}$

Theorem: If [Equation] is [Equation] -smooth and strongly convex with parameter [Equation] then using [Equation] ,

1. [Equation] ,
2. [Equation] .

- [Equation] is referred to as the condition number of [Equation] .
- In many practical problems we do not know [Equation] and [Equation] . In these cases we may use an upper bound on [Equation] and a lower bound on [Equation] .

Comparison (objective: [Equation])

[Equation] (convex)	Number of iterations ([Equation])
Lipschitz	[Equation]
Smooth (Lipschitz gradient)	[Equation]
Smooth and strongly convex	[Equation]

Armijo's rule (with steepest descent direction):
Initialize: [Equation]
Iterate: for [Equation] where [Equation]

- Backtracking line search (Armijo's rule): Fix [Equation] and [Equation] .
  Given [Equation] ,
  Initialize: [Equation]
  While [Equation]
  [Equation]
  End
  [Equation]

Lemma: If [Equation] is [Equation] -smooth then step-size adjustment is guaranteed to terminate before [Equation] goes below [Equation] . So, [Equation] .

Theorem: If [Equation] is [Equation] -smooth then [Equation] .

Theorem: If [Equation] is [Equation] -smooth and strongly convex with parameter [Equation] and [Equation] then
[Equation] .

Newton's method
Initialize: [Equation]
Iterate: for [Equation] (assuming that [Equation] is invertible)

- This can be seen as a gradient method with [Equation] .
- Motivation: convergence will be faster by using second derivatives.

Theorem: If [Equation] is invertible and [Equation] is Lipschitz continuous with a parameter [Equation] in the neighbourhood of [Equation] then
[Equation]
provided [Equation] is close enough to [Equation] . ([Equation] denotes the smallest eigenvalue modulus of [Equation]).

- If [Equation] for all [Equation] and [Equation] s are close to [Equation] then they converges to [Equation] .
- Newton's method converges faster near [Equation] than the earlier gradient methods (order-2 convergence as opposed to order-1 convergence).

Modified Newton's method
Fix: [Equation]
Initialize: [Equation]
Iterate: for [Equation]
    If [Equation] is invertible and [Equation]
        [Equation]
Else
    find [Equation] such that [Equation]
    Set [Equation] and [Equation] according to Armijo's rule
        [Equation]

end